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Inventor(s) Okamura; Yuko et al.

Skin stimulation brush

Abstract

A skin stimulation brush pin includes an elongated tubular body, a base bottom member, an electrode member, and a metal shaft. The elongated tubular body has a proximal end portion. The tubular body includes a radially contracted portion on a distal end side of the tubular body. The base bottom member is made of a metal to allow conduction of electricity from an electric circuit. The base bottom member is at the proximal end portion of the elongated tubular body. The electrode member is connected to an opening of the tubular body on the distal end side thereof and has an exposed distal end, the radially contracted portion of the tubular body radially protruding toward the electrode member. The metal shaft is in a tubular inner portion of the tubular body to couple the base bottom member and the electrode member.

Inventors: **Okamura; Yuko (Osaka, JP), Hosokawa; Takashi (Osaka, JP), Matsuoka; Manami (Osaka, JP), Aramoto; Fujio (Osaka, JP)**

Applicant: **G.M CORPORATION CO., LTD.** (Osaka, JP); **KINJO RUBBER CO., LTD.** (Yao, JP)

Family ID: **78509597**

Assignee: **G.M CORPORATION CO., LTD. (Osaka, JP)**

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Primary Examiner: Stanis; Timothy A

Attorney, Agent or Firm: Modal PLLC

Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a skin stimulation brush that electrically stimulates skin.

BACKGROUND ART

(2) There is known a related-art brush that is made of, for example, a resin. The related-art brush includes a handle, a brush base, and a plurality of bristles. The brush base is provided at a distal end of the handle. The plurality of bristles are provided upright on the brush base.

(3) Further, there is known a hair growth brush having the following configuration as a brush with which a scalp can be massaged. Specifically, the hair growth brush includes a container that stores a solution such as a hair growth product, bristles that are connected to the container to discharge the solution onto a scalp, a power source for a current that electrically stimulates a living body, and bristles that are connected to the power source to allow the current to flow to the scalp.

BRIEF SUMMARY

(4) In the related art, however, the bristles that allow a current to flow to skin are exposed on a surface. Thus, there is a problem in that the bristles may swell and degrade due to a drug such as a hairdressing product or sweat and thus electroconductivity is lowered with use.

(5) In order to solve the above-mentioned problem, according to the present invention, there is provided a skin stimulation brush, including: a housing that forms a handle and a brush base provided to one end of the handle; an electric circuit accommodated in the housing; and a plurality of skin stimulation brush pins provided to the brush base in a protruding manner, wherein each of the skin stimulation brush pins includes: a base bottom member made of a metal, which is connected to the electric circuit so as to allow conduction of electricity from the electric circuit; a body member being an elongated tubular body, which has a proximal end portion at which the base bottom member is disposed; an electrode member, which is connected to an opening of the body member on a distal end side, and has a distal end being exposed; and a metal shaft inserted into a tubular inner portion of the body member with an air gap, which is configured to couple the base bottom member and the electrode member.

(6) According to the above-mentioned configuration, the metal shaft that applies a current to the electrode member is inserted into the tubular inner portion of the body member. Thus, the metal shaft is not brought into direct contact with sebum, water, shampoo, a cosmetic product, or the like, which adheres to skin or body hair. As a result, corrosion of the metal shaft serving as a path for a current from the electric circuit to the electrode member can be suppressed, and lowering of electroconductivity can be suppressed for a long period of time.

(7) Further, the electrode member may have corrosion resistance.

(8) In this case, the electrode member, which is brought into direct contact with skin, has corrosion resistance. Thus, even when sebum, water, shampoo, a cosmetic product, or the like adheres to the electrode member, the electrode member is less liable to rust. As a result, corrosion of the electrode member is suppressed, and lowering of electroconductivity of the electrode member can be suppressed.

(9) Further, the electrode member may be made of stainless steel. Further, copper, aluminum, silver, gold, or gold may be used.

(10) Further, the metal shaft may have pliability that allows bending in a radial direction, and the body member may have elasticity that allows bending in a radial direction.

(11) A “radial direction” refers to a radial direction when it is assumed that a length direction of the skin stimulation brush pin is an axial direction. However, this description does not limit a shape of the metal shaft or the body member to a columnar shape or a cylindrical shape.

(12) In this case, the body member and the metal shaft inserted into the tubular inner portion of the body member can be bent in the radial direction. Thus, the skin stimulation brush pins can be bent in the radial direction in accordance with a force applied from skin to the skin stimulation brush pins during the use of the skin stimulation brush pins. Thus, the skin stimulation brush pins can

achieve a massaging effect produced by elasticity from a bent state achieved along with pressure on skin. Further, in combination with the above-mentioned configuration, reliable electroconductivity can be provided by the metal shaft. In addition, an effect of suppressing lowering of electroconductivity with elapse of time, which is caused by the contact with, for example, sebum, water, shampoo, or a cosmetic product, can be produced.
(13) Further, it is preferred that the metal shaft be formed of a spring. An axially elongated spring is preferred as the spring. More specifically, suitable examples of the spring include a leaf spring and a coil spring. The metal shaft is a coil spring that urges the electrode member toward a distal end side. When a force acting toward a proximal end side is applied to the electrode member, the electrode member may be thrust toward the proximal end side against an urging force of the coil spring.
(14) In this case, the electrode member can be thrust toward the proximal end side against the urging force of the coil spring. Thus, when the skin stimulation brush pin is brought into contact with skin, the electrode member is thrust toward the proximal end side by a pressure generated as a result of the contact with the skin. Further, the urging force of the coil spring is applied to the electrode member toward the distal end side under a state in which the electrode member is thrust toward the proximal end side. As a result, when the electrode member presses skin, an impact generated by the contact between the electrode member and the skin is alleviated. In this manner, the degree of force applied by a user can easily be adjusted.
(15) Further, the body member may have flexibility in the length direction, and when the electrode member is thrust toward a proximal end side, the body member may be compressed in the length direction.
(16) In this case, when the electrode member is thrust toward the proximal end side, the body member is compressed in the length direction. Further, the body member has flexibility. Thus, when the electrode member is thrust toward the proximal end side, not only the urging force of the coil spring but also a restoring force of the body member is applied to the distal end side. As a result, even when the electrode member strongly presses skin, the skin of the user of the skin stimulation brush pins is not unexpectedly damaged. Further, the urging force and the restoring force can provide pressure feeling with comfortable stimulation to the user while ensuring reliable contact between the electrode member and the skin.
(17) Further, the body member may have water resistance.
(18) In this case, when the body member has water resistance, intrusion of water into the tubular inner portion of the body member can be suppressed. Thus, the contact of water with the metal shaft that is inserted into the tubular inner portion can be suppressed. As a result, corrosion of the metal shaft can be suppressed, and thus lowering of electroconductivity of the metal shaft can be suppressed.
(19) Further, the body member may be made of a silicone rubber.
(20) In this case, the silicone rubber has water resistance and is safe to a human body. Thus, the skin stimulation brush pins can be used without anxiety.
(21) Further, the electric circuit may be configured to be capable of applying a low-frequency current to the electrode member from the base bottom member via the metal shaft.
(22) In this case, the low-frequency current applied to the base bottom member flows through the metal shaft to cause conduction of electricity through the electrode member. Thus, when the electrode member conducting electricity is brought into contact with skin, the low-frequency current is applied to the skin. Further, the low-frequency current has a property of acting on the vicinity of a skin surface. As a result, electric stimulation can be applied particularly to the vicinity of a skin surface as a target.
(23) According to the present invention, the metal shaft is not brought into direct contact with sebum, water, shampoo, a cosmetic product, or the like, which adheres to skin or body hair. Thus, the corrosion of the metal shaft can be suppressed. Accordingly, lowering of electroconductivity of

the metal shaft can be suppressed. As a result, a skin stimulation brush with suppression of lowering of electroconductivity can be provided. User's need to maintain intensity of the electric stimulation can also be satisfied.
(24) Further, when the body member is configured to have the tubular body with elasticity in the radial direction and flexibility in the length direction, strong pressure applied by the electrode member onto the skin of the user of the skin stimulation brush does not unexpectedly damage the skin. Further, the urging force and the restoring force can provide pressure feeling with comfortable stimulation to the user while ensuring reliable contact between the electrode member and the skin.
(25) Further, according to the present invention, the low-frequency current applied from the electrode member to skin can increase blood flow in the skin to activate the metabolism of cells. As a result, when the skin of an area having hair is stimulated, hair loss can be suppressed. Further, when a facial skin is stimulated, a rapid skin lift-up effect can be exerted. Thus, a skin condition before makeup is improved, and makeup sits better.
(26) Further, the skin lift-up effect can also contribute to a firming effect on a skin other than a facial skin, and can suppress sagging of the skin.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** is a left side view of a skin stimulation brush (**1**).
(2) FIG. **2** is a front view of the skin stimulation brush (**1**).
(3) FIG. **3** is an enlarged view of a skin stimulation brush pin (**10**) of the skin stimulation brush (**1**) according to the present invention.
(4) FIG. **4** is an enlarged sectional view of the skin stimulation brush pin (**10**) of the skin stimulation brush (**1**) according to the present invention.
DETAILED DESCRIPTION
(5) Now, an embodiment of the present invention is described in detail with reference to the accompanying drawings. Components having substantially the same functions and configurations are denoted by the same reference symbols in this specification and the drawings, and an overlapping description thereof is herein omitted. Moreover, the embodiment given below shows an example of the present invention. Therefore, the technical scope of the present invention is not limited to this embodiment.
(6) First, with reference to FIG. **1** and FIG. **2**, an example of a configuration of a skin stimulation brush **1** according to this embodiment is described. FIG. **1** is a left side view of the skin stimulation brush **1**. FIG. **2** is a front view of the skin stimulation brush **1**.
(7) The skin stimulation brush **1** includes a handle **3** and a brush base **4** provided to one end of the handle **3**. The handle **3** and the brush base **4** are formed of a housing **2**. A plurality of skin stimulation brush pins **10** are provided to the brush base **4** in a protruding manner. An electric circuit **6** is accommodated in the housing **2**.
(8) The handle **3** has an elliptic cylindrical shape with a diameter that increases from a lower end toward an intermediate portion and then decreases toward an upper end. One end of the handle **3** is connected to the brush base **4**. The handle **3** is a portion to be gripped by a user. The handle **3** is made of, for example, a resin. A peripheral surface of the handle **3** is formed as the housing **2**.
(9) The brush base **4** is formed in a block-like shape with a protruding back surface. One end of the brush base **4** is connected to one end of the handle **3**. The plurality of skin stimulation brush pins **10** are provided to the brush base **4** in a protruding manner. The brush base **4** may be made of, for example, a resin. A peripheral surface of the brush base **4** is formed as the housing **2**. A plurality of brush-base holes **5** into which the skin stimulation brush pins **10** are inserted may be formed in the brush base **4**. The brush-base holes **5** are formed as small holes passing through the housing **2**. The

brush-base holes **5** are formed in one side surface of the brush base **4** in a dispersed manner. In one example described in this embodiment, the brush base **4** is formed in a block-like shape with a protruding back surface, and has the brush-base holes **5** formed therein. However, a shape of the brush base **4** is not limited to thereto. The brush base **4** may have any shape as long as the plurality of skin stimulation brush pins **10** can be provided thereto in a protruding manner. For example, body portions **30** of the skin stimulation brush pins **10** and the brush base **4** may be formed integrally instead of forming the brush-base holes **5**.

(10) The handle **3** and the brush base **4** are formed by joining two housings **2** divided into a front part and a rear part. The handle **3** and the brush base **4** may be formed integrally, or the handle **3** and the brush base **4**, which are independent of each other, may be joined together at one end of the handle **3** and one end of the brush base **4**.

(11) The housing **2** forms the handle **3** and the brush base **4**. The electric circuit **6** is accommodated inside the housing **2**. The housing **2** may accommodate not only the electric circuit **6** but also a battery or a cell, which is connected to the electric circuit **6**. A connector to be used for charging may be provided to the housing **2**. The housing **2** may accommodate a cell case.

(12) The electric circuit **6** is accommodated in the housing **2**. The electric circuit **6** may be formed of a printed circuit board. The electric circuit **6** has a plurality of through holes **7** formed therein. The through holes **7** are formed at positions corresponding to the brush-base holes **5** when the electric circuit **6** is accommodated in the housing **2**. A battery or a cell may be connected to the electric circuit **6**. The electric circuit **6** may be configured to be capable of applying a low-frequency current to base bottom members **20**. For example, publicly known various low-frequency current generators or low-frequency modulators may be provided to the electric circuit **6**.

(13) Next, with reference to FIG. **3** and FIG. **4**, an example of the skin stimulation brush pin **10** of the skin stimulation brush **1** according to this embodiment is described. FIG. **3** is an enlarged view of the skin stimulation brush pin **10** of the skin stimulation brush **1** according to the present invention. FIG. **4** is an enlarged sectional view of the skin stimulation brush pin **10** of the skin stimulation brush **1** according to the present invention.

(14) The skin stimulation brush pin **10** includes the base bottom member **20** made of a metal, which can be connected to the electric circuit **6** so as to allow conduction of electricity from the electric circuit **6**. The skin stimulation brush pin **10** includes the body member **30**, which is an elongated tubular body. The base bottom member **20** is disposed at a proximal end portion of the body member **30**. The skin stimulation brush pin **10** includes an electrode member **50**. The electrode member **50** is connected to a distal-end opening portion **34** being an opening formed on a distal end side of the body member **30**, and has a distal end being exposed. The skin stimulation brush pin **10** includes a metal shaft **40**. The metal shaft **40** is inserted into the tubular inner portion of the body member **30** to thereby couple the base bottom member **20** and the electrode member **50** to each other.

(15) The base bottom member **20** is formed as a rotary body having an axis extending in a length direction of the skin stimulation brush pin **10** as a center. The base bottom member **20** is formed to be solid. A pin **21** having an elongated shape is formed in a protruding manner at a proximal end portion of the base bottom member **20**. The pin **21** is provided so as to protrude from a main body of the base bottom member **20** toward a proximal end side. A base bottom flange **22** is formed on a side periphery of the main body portion of the base bottom member **20**. A base bottom protruding portion **23** having a columnar shape is formed in a protruding manner at a distal end portion of the base bottom member **20**. The base bottom member **20** is made of a metal. The base bottom member **20** is made of stainless steel.

(16) As illustrated in FIG. **3**, the base bottom member **20** is connected to the electric circuit **6** so as to allow conduction of electricity from the electric circuit **6**. The pin **21** of the base bottom member **20** is inserted into the through hole **7**. When the pin **21** is inserted into the through hole **7**, the pin **21** and the through hole **7** are brought into contact with each other to thereby connect the base

bottom member **20** to the electric circuit **6**. In one example described in this embodiment, the base bottom member **20** is connected to the electric circuit **6** through the contact of the pin **21** with the through hole **7**. However, a connection method to the electric circuit **6** is not limited thereto. Various methods can be used as long as the base bottom member **20** is connected to the electric circuit **6**. For example, an electroconductive cord connected to the electric circuit **6** may be connected to the base bottom member **20**.

(17) As illustrated in FIG. **3** and FIG. **4**, in one example described in this embodiment, the base bottom member **20** is formed as an independent component. However, a configuration of the base bottom member **20** is not limited thereto. Various methods can be used as long as conduction of electricity through the metal shaft **40** is allowed. For example, the base bottom member **20** may be formed integrally with a proximal end portion of the metal shaft **40** as a part of the metal shaft **40**.

(18) The body member **30** is an elongated tubular body, and has a proximal end portion at which the base bottom member **20** is disposed. The body member **30** is formed in a cylindrical shape. The body member **30** is formed with a diameter decreasing toward its distal end. The body member **30** has an outer diameter gradually decreasing toward the distal end and an inner diameter correspondingly gradually decreasing toward the distal end. The body member **30** has a proximal-end opening portion **31** and the distal-end opening portion **34**. The proximal-end opening portion **31** is an opening on the proximal end side, and the distal-end opening portion **34** is an opening on the distal end side. The body member **30** has a body stopper portion **32** having a flange-like shape. The body stopper portion **32** is formed on an outer peripheral surface of the proximal end portion of the body member **30**. The body member **30** has a radially contracted portion **33** at which an inner diameter is reduced. The radially contracted portion **33** forms a level difference on an inner surface of the distal end portion. An upper half part of the base bottom member **20** is inserted and fitted into the proximal-end-side opening portion of the body member **30** toward the distal end side.

(19) An edge of the proximal-end opening portion **31** at the proximal end of the body member **30** is in contact with a surface of the base bottom flange **22** of the base bottom member **20** on the distal end side. The body stopper portion **32** of the body member **30** is in contact with an inner surface of the housing **2**. The body stopper portion **32** of the body member **30** is caught by an edge of the brush-base hole **5** of the housing **2** so as not to be removed from the brush-base hole **5**. The edge of the proximal-end opening portion **31** at the proximal end of the body member **30** is in contact with the surface of the base bottom flange **22** of the base bottom member **20** on the distal end side, while a surface of the body stopper portion **32** on the distal end side is in contact with the inner surface of the housing **2**. Thus, the proximal end portion of the body member **30** is fixed to the brush base **4**.

(20) It is preferred that the body member **30** have elasticity that allows bending in a radial direction with respect to an axis extending in the length direction of the skin stimulation brush pin **10**. Further, it is more preferred that the body member **30** be made of a silicone rubber.

(21) The body member **30** may have flexibility in the length direction. Further, it is preferred that the body member **30** have flexibility in the length direction and be formed so as to be compressible in the length direction when the electrode member **50** is thrust toward the proximal end side.

(22) It is preferred that the body member **30** have water resistance.

(23) In one example described in this embodiment, the body member **30** is formed in a cylindrical shape. However, the shape of the body member **30** is not limited thereto. The body member **30** may be formed as an elongated tubular body having various shapes. A sectional shape of the body member **30** may be, for example, circular, elliptical, rectangular, or pentagonal.

(24) The proximal end portion of the body member **30** is disposed without leaving a gap from the brush base **4**. Thus, the stopper portion **32** of the body member **30** is in contact with the inner surface of the housing **2** to thereby achieve sealing.

(25) The base bottom member **20** is disposed at the proximal end portion of the body member **30** without leaving a gap. The base bottom member **20** is fitted into the proximal-end opening portion **31** of the body member **30** so as to be in close contact therewith.

(26) The metal shaft **40** is inserted into the tubular inner portion of the body member **30** to thereby couple the base bottom member **20** and the electrode member **50** to each other. A proximal end of the metal shaft **40** is in contact with the base bottom member **20**. A distal end of the metal shaft **40** is in contact with a shaft portion **51** of the electrode member **50**. The distal end portion of the metal shaft **40** is fixed to the shaft portion **51** of the electrode member **50** with an adhesive **60** while being in contact with the shaft portion **51**. The proximal end portion of the metal shaft **40** is fitted over the base bottom protruding portion **23** of the base bottom member **20**. The metal shaft **40** is compressed in the length direction by the base bottom member **20** that is fitted into the proximal end of the body member **30** toward the distal end side.
(27) It is preferred that the metal shaft **40** have pliability that allows bending in a radial direction with respect to an axis extending in the length direction of the skin stimulation brush pin **10**. It is more preferred that the metal shaft **40** be formed of a coil spring that urges the electrode member **50** toward the distal end side. It is preferred that the coil spring, which corresponds to the metal shaft **40**, constantly urge the electrode member **50** toward the distal end side. However, even in a case in which the coil spring does not constantly urge the electrode member **50** toward the distal end side, the coil spring is only required to urge the electrode member **50** toward the distal end side when the electrode member **50** is thrust toward the proximal end side. It is preferred that the coil spring be held and compressed between the base bottom member **20** and the electrode member **50**.
(28) In one example described in this embodiment, the metal shaft **40** is formed of a coil spring. However, the metal shaft **40** is not limited to a coil spring. The metal shaft **40** may be formed in various shapes as long as the base bottom member **20** and the electrode member **50** can be coupled to each other to allow conduction of electricity through the electrode member **50**. The metal shaft **40** can be formed, for example, in a columnar body, a wire body, or a tubular body.
(29) The electrode member **50** is connected to the distal end-side opening of the body member **30** and has a distal end being exposed. The electrode member **50** is formed as a solid block body. The electrode member **50** includes the shaft portion **51** having an elongated shape, which extends from an intermediate portion to a proximal end of the electrode member **50**. An electrode stopper portion **52** having a flange-like shape is formed on a peripheral surface of an intermediate portion of the shaft portion **51**. The shaft portion **51** of the electrode member **50** is fitted into the distal-end opening portion **34** of the body member **30**. The electrode stopper portion **52** is engaged with the level difference formed at a proximal end of the radially contracted portion **33** of the body member **30**. A proximal end portion of the shaft portion **51** is in contact with the distal end of the metal shaft **40**. A proximal end portion of the electrode member **50** is fixed to the distal end of the metal shaft **40** with the adhesive **60** while being in contact with the distal end of the metal shaft **40**. The electrode member **50** is urged toward the distal end side by the metal shaft **40**, which is a coil spring. The distal end of the electrode member **50** has a curved surface.
(30) The electrode member **50** may have corrosion resistance. It is more preferred that the electrode member **50** be made of stainless steel. Besides stainless steel, a metal containing chromium or a conductor that has been subjected to an anti-corrosive process such as plating is preferred for the electrode member **50**. Further, copper, aluminum, silver, gold, or gold can also be used for the electrode member **50**.
(31) Further, the electrode member **50** is connected to the distal-end opening portion **34**, which is the opening of the body member **30** on the distal end side, without leaving a gap. The electrode member **50** is connected to the distal-end opening portion **34** of the body member **30** while being in close contact therewith.
(32) The adhesive **60** is supplied into the tubular inner portion of the body member **30**, which surrounds the shaft portion **51** of the electrode member **50**. The adhesive **60** fixes the proximal end portion of the electrode member **50** and the distal end of the metal shaft **40** under a state in which the proximal end portion and the distal end are in contact with each other. Various adhesives **60** such as, for example, a room temperature vulcanizing (RTV) rubber can be used as the adhesive **60**.

Further, it is preferred that the adhesive **60** be an electroconductive adhesive.
(33) Now, actions and effects of the skin stimulation brush **1** are described in accordance with a procedure of use in one example of the skin stimulation brush **1** according to this embodiment.
(34) First, a user of the skin stimulation brush **1** turns on a power source (not shown) for the skin stimulation brush **1**, which is provided to the housing **2**.
(35) When the skin stimulation brush **1** described in one example of this embodiment is turned on, a current flows through the electric circuit **6**. Further, the base bottom members **20** are connected to the electric circuit **6** so as to allow conduction of electricity from the electric circuit **6**. More specifically, the base bottom members **20** are inserted into the through holes **7** while the pins **21** are in contact with the through holes **7**. Thus, the current, which flows through the electric circuit **6**, causes conduction of electricity through the base bottom members **20**. Next, the metal shafts **40** are in contact with the base bottom members **20** conducting electricity. As a result, the current flows from the base bottom members **20** to the metal shafts **40**. Further, the metal shafts **40** are in contact with the electrode members **50**. Thus, the current that has flowed through the metal shafts **40** flows to the electrode members **50**. Thus, the current is caused to flow through the electrode members **50**.
(36) Further, in the skin stimulation brush **1** described in one example of this embodiment, when the electric circuit **6** is configured to be capable of applying a low-frequency current to the base bottom members **20**, for example, when the electric circuit **6** includes a low-frequency current generator or a low-frequency modulator, a low-frequency current is applied to the base bottom members **20**. Thus, the low-frequency current flows from the base bottom members **20** via the metal shafts **40** to the electrode members **50**.
(37) Next, a massage is given while the skin stimulation brush **1** including the electrode members **50** conducting electricity is held in contact with skin. In this case, the electrode members **50** having the distal ends exposed from the body members **30** are brought into contact with skin to apply electric stimulation and pressure stimulation to the skin.
(38) As skin, skin all over the body, which includes a scalp, may be a target. For example, sebum, water, shampoo, or a cosmetic product may adhere to a scalp and skin all over the body, or body hair growing therefrom. These liquids may degrade a related-art skin stimulation brush to lower its electroconductivity. Specifically, some bristles used for related-art skin stimulation brushes are made of a metal or an electroconductive resin. When skin is to be stimulated with use of these bristles, the following problems may arise. When skin is stimulated with use of bristles made of a metal, there is a problem in that, for example, water adhering to skin may corrode a metal of the bristles to thereby lower electroconductivity. Further, when the bristles are made of an electroconductive resin, there is a problem in that the resin absorbs water or the like and degrades to thereby lower the electroconductivity of the electroconductive resin. These problems may reduce a lifetime of the skin stimulation brush itself. Thus, user's need to maintain intensity of electric stimulation over a long period of time cannot be sufficiently satisfied.
(39) Thus, in the skin stimulation brush **1** described in one example of this embodiment, the metal shaft **40**, which serves as a current path, is inserted into the tubular inner portion of the body member **30**. Thus, the metal shaft **40** is not brought into direct contact with sebum, water, shampoo, a cosmetic product, or the like, which adheres to skin or body hair. As a result, corrosion of the metal shaft **40**, which serves as a current path from the electric circuit **6** to the electrode member **50**, can be suppressed, and thus lowering of electroconductivity can be suppressed for a long period of time. Further, this allows intensity of electric stimulation to skin to be maintained over a long period of time. Thus, a lifetime not only as the metal shaft **40** but also as the skin stimulation brush **1** as a whole can be extended.
(40) Further, nonuniformity in degradation, that is, degradation in electroconductivity of only some of the plurality of skin stimulation brush pins **10** mounted to the skin stimulation brush **1** described in one example of this embodiment due to adhesion of, for example, a cosmetic product, can be prevented.

(41) Further, when the body member **30** of the skin stimulation brush **1** described in one example of this embodiment has water resistance, the body member **30** into which the metal shaft **40** is inserted does not easily allow water to pass into the tubular inner portion. Thus, the metal shaft **40** is not brought into contact with liquid. Accordingly, a more remarkable effect of suppressing the corrosion of the metal shaft **40** is obtained.
(42) When the body member **30** of the skin stimulation brush **1** described in one example of this embodiment is made of a silicone rubber, excellent water resistance can be obtained owing to physical properties of the silicone rubber to thereby prevent the corrosion of the metal shaft **40**. Further, when the body member **30** is made of a silicone rubber, the body member **30** has elasticity that allows bending in the radial direction. Further, when the body member **30** is made of a silicone rubber, the body member **30** also has flexibility in the length direction.
(43) When the electrode members **50** of the skin stimulation brush **1** described in one example of this embodiment has corrosion resistance, the corrosion of the electrode members **50** can be suppressed. Thus, lowering of electroconductivity of the electrode members **50** can be suppressed.
(44) Further, when the electrode member **50** of the skin stimulation brush **1** described in one example of this embodiment is made of stainless steel, the electrode member **50** has corrosion resistance. Thus, the corrosion of the electrode member **50** can be suppressed to thereby suppress lowering of electroconductivity. In addition, when the electrode member **50** is made of stainless steel, the corrosion of the electrode member **50** can be suppressed over a longer period of time and lowering of electroconductivity can be suppressed owing to a characteristic that a passivation film is regenerated. The passivation film of stainless steel is extremely thin, and thus does not inhibit an electric stimulation effect on skin. Further, when the electrode member **50** is made of stainless steel, processibility and strength of the electrode member **50** can easily be improved.
(45) Further, when the body member **30** and the electrode member **50** of the skin stimulation brush **1** described in one example of this embodiment are connected to each other without leaving a gap, intrusion of liquid or moisture through a gap between the body member **30** and the electrode member **50** into the inside of the tubular body of the body member **30** can be suppressed. As a result, the corrosion of the metal shaft **40** can be suppressed to thereby suppress lowering of electro conductivity.
(46) Further, when the brush base **4** and the body member **30** of the skin stimulation brush **1** described in one example of this embodiment are connected to each other without leaving a gap, intrusion of liquid or moisture through a gap between the body member **30** and the brush base **4** into the inside of the tubular body of the body member **30** can be suppressed. As a result, the corrosion of the metal shaft **40** or the base bottom member **20** can be suppressed to thereby suppress lowering of electro conductivity.
(47) When the body member **30** and the electrode member **50** are in close contact with each other at their connecting portion without leaving a gap and the body member **30** and the brush base **4** are in close contact with each other at their connecting portion without leaving a gap, liquid or moisture can be prevented from intruding into the body member **30**. In particular, lowering of electroconductivity of the skin stimulation brush **1** can be prevented, and a lifetime of the skin stimulation brush **1** can be extended. Further, this configuration can easily be achieved by using the body member **30** having elasticity such as a silicone rubber.
(48) When the distal end of the electrode member **50** of the skin stimulation brush **1** described in one example of this embodiment has a curved surface, contact of the electrode member **50** with skin can provide comfortable feeling.
(49) Next, the skin stimulation brush pins **10** of the skin stimulation brush **1** are brought into contact with skin obliquely with respect to a skin surface.
(50) In this case, when the metal shaft **40** of the skin stimulation brush **1** described in one example of this embodiment has pliability that allows bending in the radial direction and the body member **30** has elasticity that allows bending in the radial direction, the body member **30** and the metal shaft

40 can be integrally bent in the radial direction. As a result, skin is pressed with the restoring force generated due to the elasticity of the body members **30** to thereby provide an excellent massaging effect.
(51) Further, when the metal shaft **40** of the skin stimulation brush **1** described in one example of this embodiment has a second elasticity, a second restoring force generated due to the second elasticity of the metal shaft **40** acts in addition to the restoring force generated due to the elasticity of the body member **30**. As a result, the skin can be pressed more strongly. Thus, a further excellent massaging effect can be obtained while electric stimulation is applied to skin. Further, for example, when the metal shaft **40** is formed of a spring, the above-mentioned second elasticity can be obtained. An axially elongated spring is preferred as the spring. More specifically, suitable examples of the spring include a leaf spring and a coil spring. In particular, a coil spring is preferred because the coil spring enables easy mounting of the electrode member **50**. More specifically, when the metal shaft **40** is a coil spring, the electrode member **50** and the metal shaft **40** can easily be connected by winding the coil spring around a shaft-shaped portion, specifically, the shaft portion **51** of the electrode member **50** inserted into the distal-end opening portion **34**. Further, as illustrated in FIG. **4**, it is preferred that the metal shaft **40** be inserted with an air gap **36** from an inner surface **35** of the body member **30**. With the air gap **36**, the metal shaft **40** can be bent without interfering with the body member **30** in a bending operation of the body member **30** in the radial direction.
(52) Next, the skin stimulation brush pins **10** of the skin stimulation brush **1** are brought into contact with skin perpendicularly from above with respect to a skin surface.
(53) In this case, when each of the metal shafts **40** of the skin stimulation brush **1** described in one example of this embodiment is formed of a coil spring, the electrode member **50** can be thrust toward the proximal end side against an urging force of the coil spring. Thus, when the electrode member **50** presses skin, an impact generated by the contact of the electrode member **50** with the skin can be alleviated. As a result, a degree of pressure exerted by the user on the skin stimulation brush **1** can easily be adjusted.
(54) Further, the urging force of the coil spring is transmitted to the distal end side of the electrode member **50** under a state in which the electrode member **50** is thrust toward the proximal end side. As a result, the electrode member **50** can press the skin to thereby achieve a further excellent massage.
(55) Further, the electrode member **50** is not limited to be thrust toward the proximal end side together with the body member **30**. For example, the electrode member **50** may be thrust toward the proximal end side while sliding with respect to the distal-end opening portion **34** of the body member **30**.
(56) Further, when the body member **30** of the skin stimulation brush **1** described in one example of this embodiment has flexibility in the length direction and the body member **30** is compressed in the length direction when the electrode member **50** is thrust toward the proximal end side, not only the urging force of the coil spring but also the restoring force toward the distal end side of the body member **30** being in a compressed state acts on the electrode member **50**. Thus, the contact between skin and the electrode members **50** can be more reliably achieved, and stronger pressure stimulation can be obtained. In the skin stimulation brush **1** described in this embodiment, when the body member **30** is compressed, the inner diameter of the tubular inner portion is increased. Thus, the degree of compression of the body member **30** can be increased. As a result, stronger pressure feeling can be obtained.
(57) Other actions and effects of the skin stimulation brush **1** described in one example of this embodiment are described.
(58) With the skin stimulation brush **1** described in one example of this embodiment, the low-frequency current applied from the electrode members **50** to skin can increase blood flow in the skin to activate the metabolism of cells. Thus, when the skin of an area having hair is stimulated,

hair loss can be suppressed. Further, when a facial skin is stimulated, a rapid skin lift-up effect can be exerted. Thus, a skin condition before makeup is improved, and thus makeup sits better.

(59) Further, the skin lift-up effect can also contribute to a firming effect on a skin other than a facial skin, and can suppress sagging of the skin.

(60) The proximal end portion of the body member **30** of the skin stimulation brush **1** described in one example of this embodiment is fixed by the inner surface of the housing **2** and a surface of the electric circuit **6**, which is opposed to the housing **2**. Thus, problems such as removal of the skin stimulation brush pins **10** from the brush-base holes **5** or forcing the skin stimulation brush pins **10** deeply into the brush-base holes **5** can be suppressed.

(61) Further, in the skin stimulation brush **1** described in one example of this embodiment, the electrode stopper portion **52** of the electrode member **50** is engaged with the radially contracted portion **33** of the body member **30**. Thus, removal of the electrode member **50** from the distal-end opening portion **34** of the body member **30** can be suppressed.

(62) Further, in the skin stimulation brush **1** described in one example of this embodiment, the coil spring is sandwiched and compressed between the base bottom member **20** and the electrode member **50**. Thus, an extension force is accumulated in the coil spring, and the contact between the coil spring, and the base bottom member **20** and the electrode member **50** can be more reliably achieved.

(63) Further, in the skin stimulation brush **1** described in one example of this embodiment, a contact portion between the metal shaft **40** and the electrode member **50** is fixed with the adhesive **60**. Further, an RTV rubber, which can be bonded to a silicone rubber, can be used as the adhesive **60**. Thus, the contact between the metal shaft **40** and the electrode member **50** can be further reliably achieved, and the removal of the electrode member **50** from the body member **30** made of a silicone rubber can be prevented.

(64) Further, when an electroconductive adhesive is used as the adhesive **60** for the skin stimulation brush **1** described in one example of this embodiment, a current, which flows through the adhesive **60**, can cause conduction of electricity through the electrode member **50** even if the metal shaft **40** and the electrode member **50** are separated from each other.

(65) Further, with a configuration in which the base bottom protruding portion **23** of the skin stimulation brush **1** described in one example of this embodiment is fitted into the proximal end portion of the coil spring, positioning at the proximal end portion of the coil spring is reliably achieved. Thus, electrical connection between the base bottom member **20** and the coil spring is reliably achieved.

(66) It should be noted that the above-mentioned actions and effects are actions and effects of the skin stimulation brush **1** described in one example of this embodiment. Thus, the technical scope of the present invention is not limited by the actions and effects of this embodiment.

REFERENCE SIGNS LIST

(67) **1** skin stimulation brush **2** housing **3** handle **4** brush base **5** brush-base hole **6** electric circuit **7** through hole **10** skin stimulation brush pin **20** base bottom member **21** pin **22** base bottom flange **23** base bottom protruding portion **30** body member **31** proximal-end opening portion **32** body stopper portion **33** radially contracted portion **34** distal-end opening portion **36** air gap **40** metal shaft **50** electrode member **51** shaft portion **52** electrode stopper portion **60** adhesive

Claims

1. A skin stimulation brush pin, comprising: an elongated tubular body having a proximal end portion, wherein the tubular body includes a radially contracted portion on a distal end side of the tubular body; a base bottom member made of a metal to allow conduction of electricity from an electric circuit, the base bottom member at the proximal end portion of the elongated tubular body; an electrode member connected to an opening of the tubular body on a distal end side thereof and

having an exposed distal end, a portion of the electrode member adjacent to a distal tip of the tubular body protruding farther in the radial direction than the opening, the radially contracted portion of the tubular body radially protruding toward the electrode member; and a metal shaft in a tubular inner portion of the tubular body to couple the base bottom member and the electrode member, the tubular inner portion including an inner surface, the metal shaft separated from the inner surface by an air gap.

2. The skin stimulation brush pin according to claim 1, wherein the metal shaft is a spring configured to urge the electrode member toward a distal end side.

3. The skin stimulation brush pin according to claim 1, wherein the metal shaft is a coil spring configured to urge the electrode member toward a distal end side.

4. The skin stimulation brush pin according to claim 1, wherein the tubular body has flexibility in the length direction, and wherein, when the electrode member is thrust toward a proximal end side, the tubular body is compressed in the length direction.

5. The skin stimulation brush pin according to claim 1, wherein the base bottom member is formed integrally with a proximal end portion of the metal shaft as a part of the metal shaft.

6. The skin stimulation brush pin according to claim 1, wherein the electrode member has corrosion resistance.

7. The skin stimulation brush pin according to claim 1, wherein the electrode member is made of stainless steel.

8. The skin stimulation brush pin according to claim 1, wherein the tubular body has water resistance.

9. The skin stimulation brush pin according to claim 1, wherein the tubular body is made of a silicone rubber.

10. The skin stimulation brush pin according to claim 3, wherein the electrode member includes a shaft portion, and the coil spring is wound around the shaft portion of the electrode member.

11. The skin stimulation brush pin according to claim 1, further comprising: an adhesive that fixes a proximal end portion of the electrode member and a distal end of the metal shaft.

12. The skin stimulation brush pin according to claim 11, wherein the adhesive is electroconductive.

13. The skin stimulation brush pin according to claim 1, wherein when the tubular body is compressed, an inner diameter of the tubular inner portion is increased.

14. A skin stimulation brush pin, comprising: an elongated tubular body having a proximal end portion and a distal end side, the tubular body including on the distal end side an opening and a distal tip, the tubular body further including on the distal end side a portion contracted in a radial direction of the skin stimulation brush pin; an electrode member connected to the opening and having an exposed distal end, a portion of the electrode member adjacent to the distal tip protruding farther in the radial direction than the opening, the radially contracted portion radially protruding toward the electrode member; urging means for urging the electrode member in a distal direction when the electrode member is thrust in a proximal direction, the urging means in a tubular inner portion of the tubular body; and conducting means for allowing conduction of electricity to the urging means, the conducting means at the proximal end portion.